COMP3006: Full-Stack Development Report 2026/27

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Links

GitHub Repository: github.com/bobbymannino/comp3006

YouTube Video: *TODO*

Introduction

The project I have decided to pursue is making a blog. The functionality of the blog is as follows:

- Anybody can create an account
- Users can edit/delete their own accounts
- Users can create/read/update/delete their own posts
- Users can like their own and other peoples posts
- Users can publish/unpublish their posts
- On each posts page there will be a live visitor count

Because I have prior knowledge in this specific field (web/API/database), I do not have a priority list for the features. I am confident I can deliver all the features to a high standard on time.

Target Audience

There is no specific demographic or user base in mind for this project, it is designed for any user. A high priority for this project is **usability** and **efficiency**. I would like the people who use this to know instantly without any training or documentation how to use the product, as well as the product feeling fast to use.

Existing Products

There are several existing products that provide similar functionality to this project. Some of the most popular ones include:

Blogger

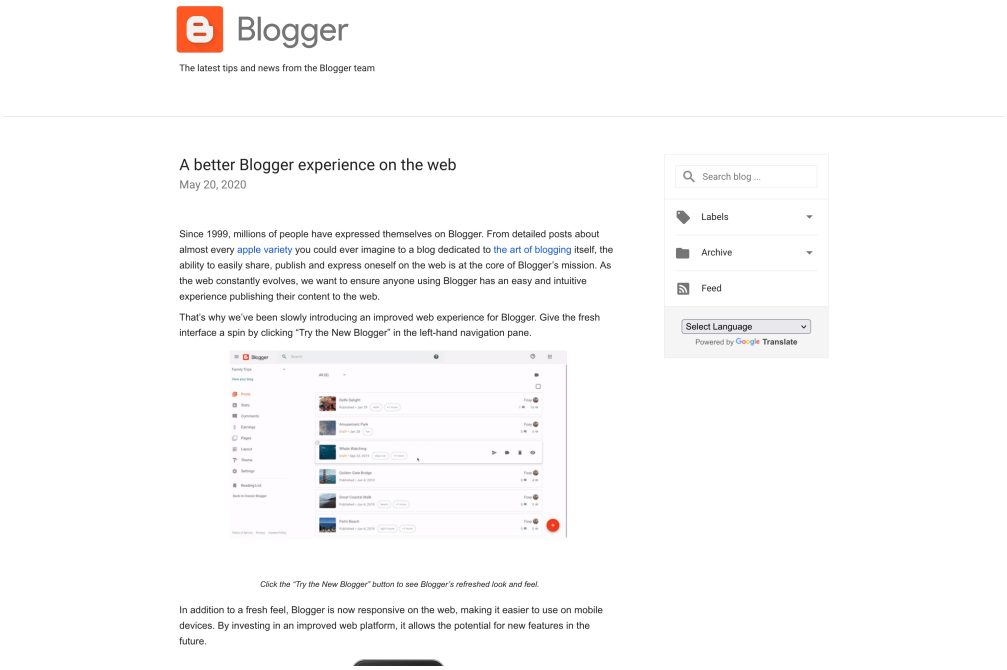


Figure 1: Blogger

I have not used Blogger in over 5 years, but when I was using it for school it got the job done. I will take inspiration from Blogger from a UX perspective, keeping things simple and easy to understand was and is Bloggers best feature.

WordPress

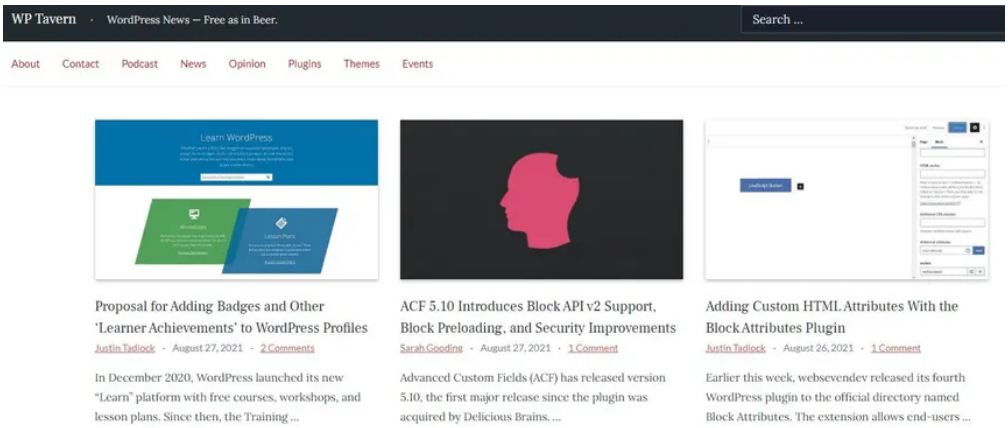


Figure 2: WordPress

I've never used WordPress before, but I know it has a massive market share online. I also know it's more than a blog; it can do nearly anything, including customizing the look. That customization is beyond this project's scope. I'll use WordPress's complexity as a warning: there are many plugins, and while they help, they can be overwhelming for someone wanting a straightforward public blog.

Architecture

Architecture Topology

The product will follow a microservices architecture. There will be 4 Docker images in this project:

- Nginx
- Database
- API
- Web UI

Each one runs independently and communicate with each other, for instance to access the database you must use the API, to use the web UI you must use the Nginx server. Putting the API and Web UI into their own containers allows you to scale them independently and update them separately. If traffic becomes API heavy you can scale those up, or if the user wants to host the API without having any UI they can do that without any changes to the API container.

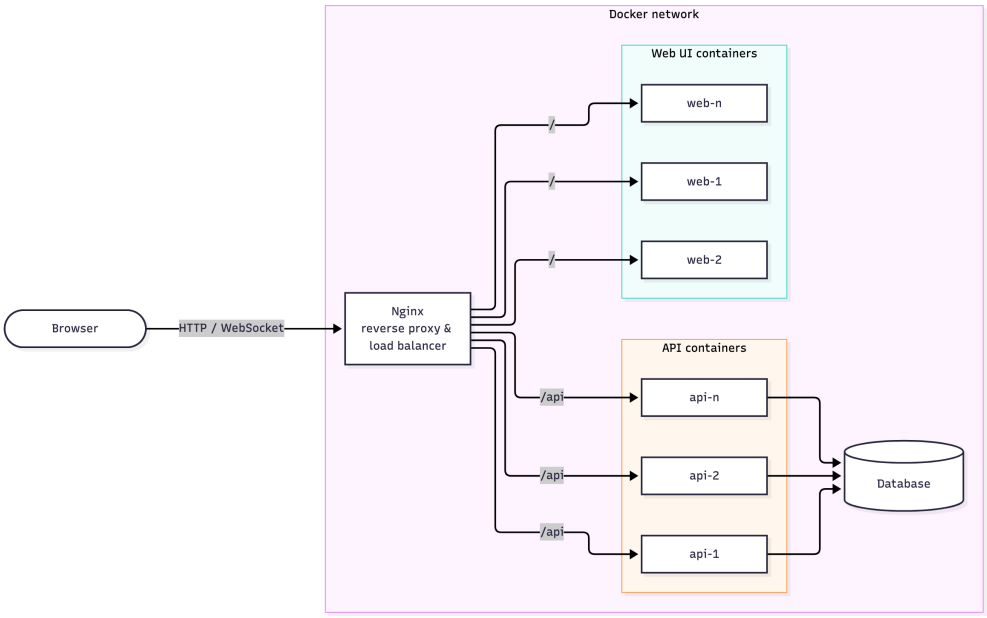


Figure 3: Architecture Topology

> **Note**
>
> Whilst it is possible to have multiple databases (read replicas) that is out of scope for this project so it will remain a single container.

Nginx

TODO

Web UI

TODO

Database

TODO

API

TODO